Data-Driven Strategies for Predicting and Mitigating Membrane Fouling in High-Risk Nuclear Wastewater Systems - 26040

Josephine Adelegan[1,2], Yogesh Bhattarai[1] , Sanjib Sharma[1], and Kimberly Jones [1,2] • [1] Department of Civil and Environmental Engineering, Howard University, Washington, DC, USA
[2] Consortium for Risk Evaluation with Stakeholder Participation (CRESP), Vanderbilt University, Nashville, TN, USA

THE PROBLEM

Hanford stores ~ 200 million liters of radioactive tank waste.

Fig 1. Hanford Radioactive Tank Waste.

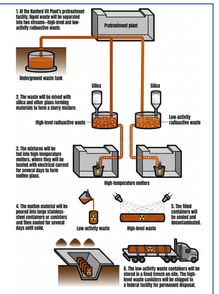


Fig 2. Hanford Vitrification Process.

Crossflow ultrafiltration(CUF) is required for pretreatment, but **membrane fouling causes persistent flux decline**, increasing cost and operational risk.

Bench-scale studies show:

- Long-term flux decay(>100 hrs)
- Sensitivity to gel concentration
- Chemistry-dependent behavior
- No sustained steady-state flux

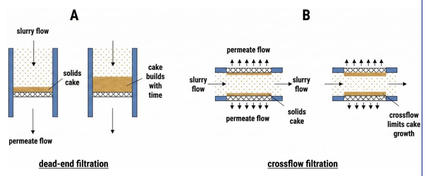


Fig 3.Dead-End and Crossflow Filtration Configurations.

Classic cake/gel models struggle to generalize across compositions.

RESEARCH OBJECTIVE

Develop a **transparent, reproducible machine learning workflow** to:

- Predict normalized permeate flux
- Quantify model uncertainty
- Identify dominant fouling drivers
- Support DOE filtration decision-making

Goal: Provide decision support **without physical modification** of filtration systems.

EXPERIMENTAL FOUNDATION

Bench-Scale Crossflow Ultrafiltration

- 8 ft sintered stainless steel tubular membrane
- Axial velocity: ~4.48 m/s
- TMP: 20 psi
- NaOH: 1, 2.5, 5 M
- Iron & aluminum simulants(gibbsite, boehmite, mixed)

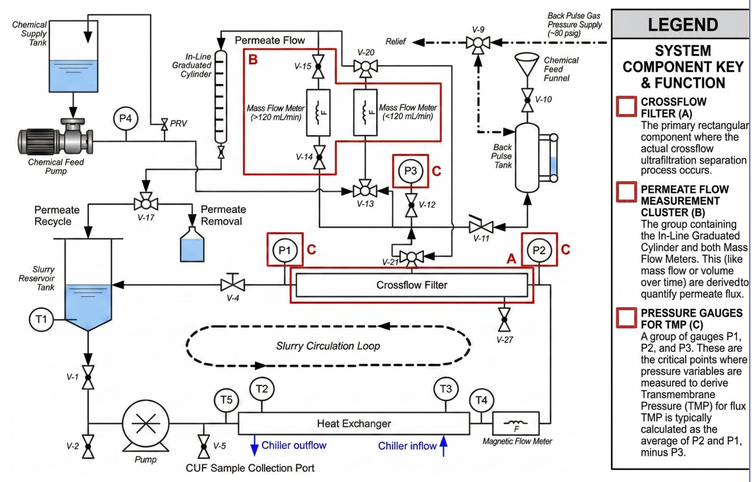


Fig 4. Simplified CUF Process & Instrumentation Diagram.

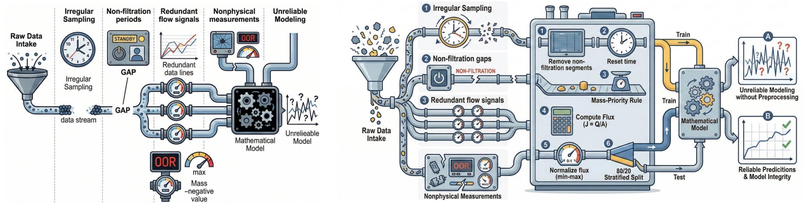


Fig 5. Raw Data Challenges & Data Processing Workflow.

RANDOM FOREST MODEL

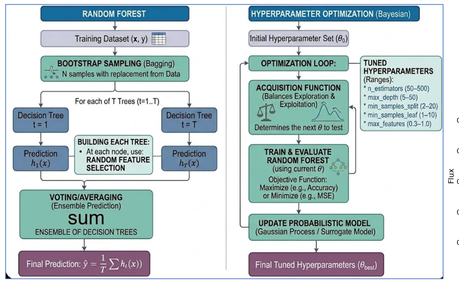
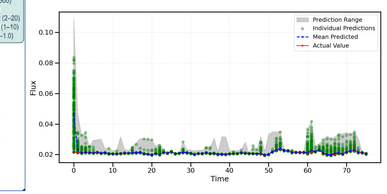


Fig 6. Time-series comparison of actual vs. predicted flux values using the Random Forest (RF) model.



FEATURE IMPORTANCE(SHAP)

- Flow rate**(strongest)**
- Filter velocity
- NaOH concentration
- TMP**(minor)**
- Temperature**(very minor)**
- Density & area**(negligible)**

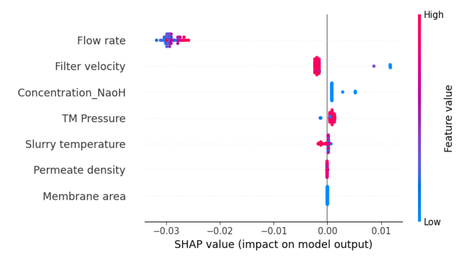
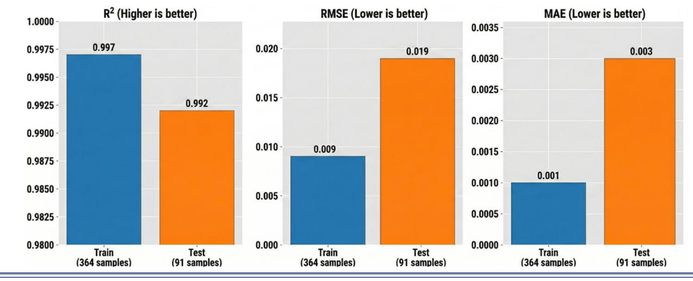


Fig 7. Feature importance for the flux prediction using Random Forest.

MODEL PERFORMANCE

Fig 8. Predictive Performance of Random Forest.



Conclusions: Settling and crossflow filtration performance are strongly governed by **solids concentration, particle size, and NaOH molarity**, with **lower NaOH concentrations enhancing settling rates and filtration flux**, while particle characteristics dictate regime behavior.

Summary: Bench-scale gravity settling, centrifugation, and crossflow filtration of iron-aluminum sludge simulants demonstrated **concentration-dependent hindered settling**, **viscosity-driven compaction trends**, and **chemistry-sensitive filtration responses**, including distinct **cake-formation and non-cake regimes**.

Takeaways: **Higher ionic strength reduces settling and filtration efficiency. Centrifuged solids provide an upper-bound gel point for filtration modeling. Aluminum phase composition critically influences sediment consolidation and membrane flux behavior.**

Recommendation: Integrate **predictive settling and filtration models** with **centrifuged solids measurements across varying chemistries** to optimize pretreatment flow paths and ensure reliable achievement of **target feed solids concentrations**.

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